

id () Function

This function returns the “identity” of an object.

The identity number is an integer which is guaranteed to be unique and constant for this object during its lifetime.

Two objects with non-overlapping lifetimes may have the same id() value.

CPython implementation detail – This is the address of the object in memory

Syntax:- id(object)

Ex:- id(lst)

type () Function

This function returns type of the object.

Syntax:- type(object)

type(a)

getsizeof() Function

This function returns the size of an object in bytes.

The object can be any type of object.

All built-in objects will return correct results, but this does not have to hold true for third-party extensions as it is implementation specific.

Only the memory consumption directly attributed to the object is accounted for, not the memory consumption of objects it refers to.

This is part of sys module so you have to import sys module before using this function.

Syntax:-

```
from sys import *  
getsizeof(object)
```

```
Ex:- from sys import *  
getsizeof(a)
```